



Introduction

ADAL: AI-Driven Analysis for Law

ADAL is an AI-powered legal research and assistance system that simplifies document analysis, citation verification, and legal drafting for Pakistani law professionals.

Team Members:

- Husnain Ali Arshad (22I-1335)
- Hasan Ali (22I-0919)
- Abdullah Azeem (22I-1186)

Supervisor: Mr. Muhammad Almas

Department: Computer Science

University: FAST–NUCES Islamabad

Project Overview

Problem

Legal research in Pakistan is manual, slow, and error-prone. Lawyers spend hours verifying citations and searching precedents.

Solution

ADAL automates legal document ingestion, OCR, citation validation, summarization, and precedent retrieval using AI and NLP models.

Goal

To provide an intelligent web-based system that enhances accuracy, efficiency, and accessibility in legal research.

Proposal Defense Comments & Feedback

Feedback Received

- Project approved with minor modifications.
- Dataset definition to be made more detailed.
- Need clear description of OCR, frontend, and backend design.

Action Taken

- Dataset finalized: Pakistani legal documents (judgments, notices, case summaries).
- OCR pipeline implemented with preprocessing.
- Frontend (React.js) and Backend (FastAPI + PostgreSQL) architectures detailed.

Functional & Non-Functional Requirements

Functional Requirements

- Document upload and OCR processing (PDF, DOCX, images).
- Citation detection and verification.
- AI-based summarization and precedent retrieval.
- User registration, authentication, and role-based access.

Non-Functional Requirements

- Analyze large documents within minutes.
- 99% uptime.
- Scalable backend for future AI model integration.
- Intuitive UI supporting dark mode and adaptive layout.

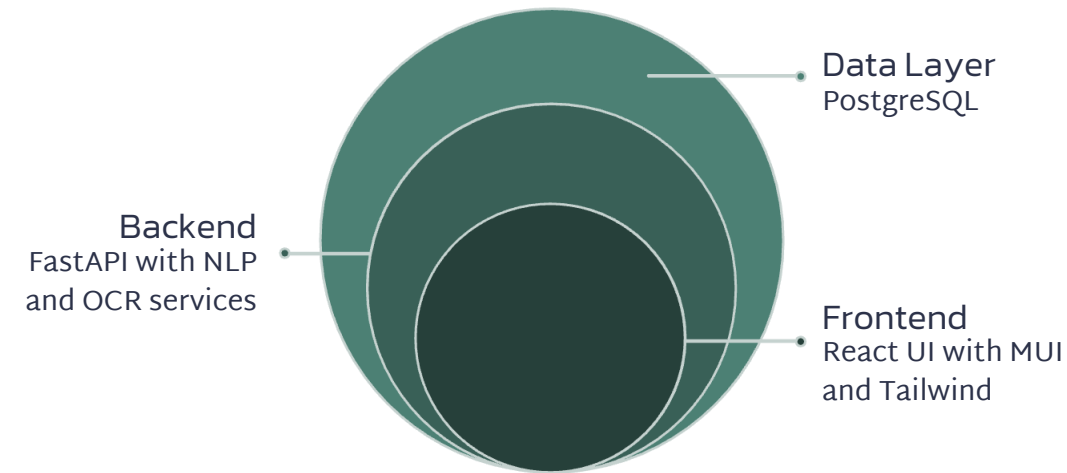
System Modules

01	02	03
Document Ingestion & Preprocessing	Citation Verification	AI Summarization & Precedent Retrieval
OCR, metadata extraction, text cleaning.	NLP-based reference validation.	Legal-BERT, RAG-based insights.
04	05	
Legal Assistant	User Interface & Reporting	
Interactive chatbot for legal queries.	Display analysis, generate reports.	

 **Current Focus (Iteration 1):** Module 1 — OCR-enabled document ingestion and preprocessing.

System Architecture

Architecture Type: Multi-tier Client–Server Architecture

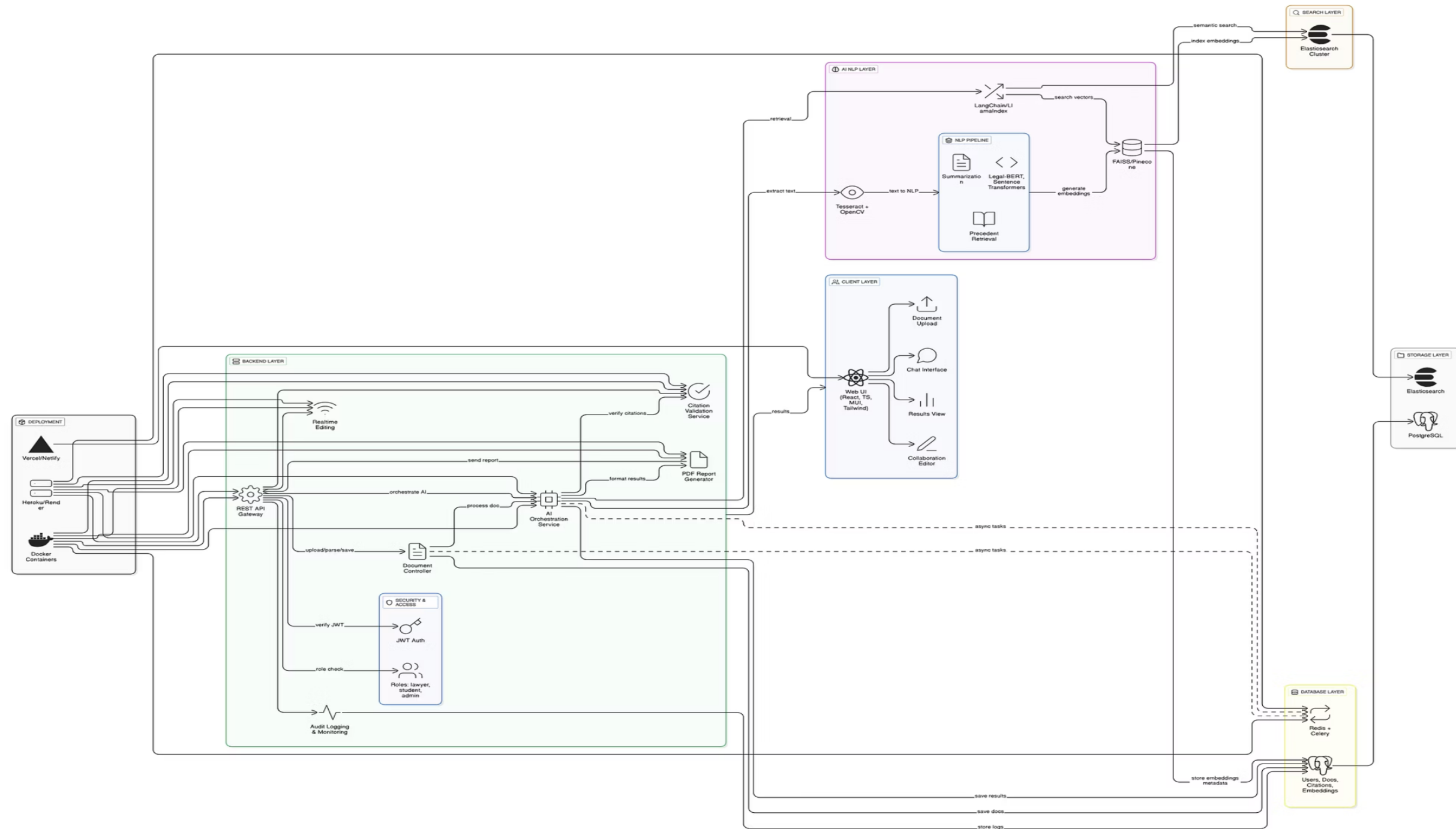


Components:

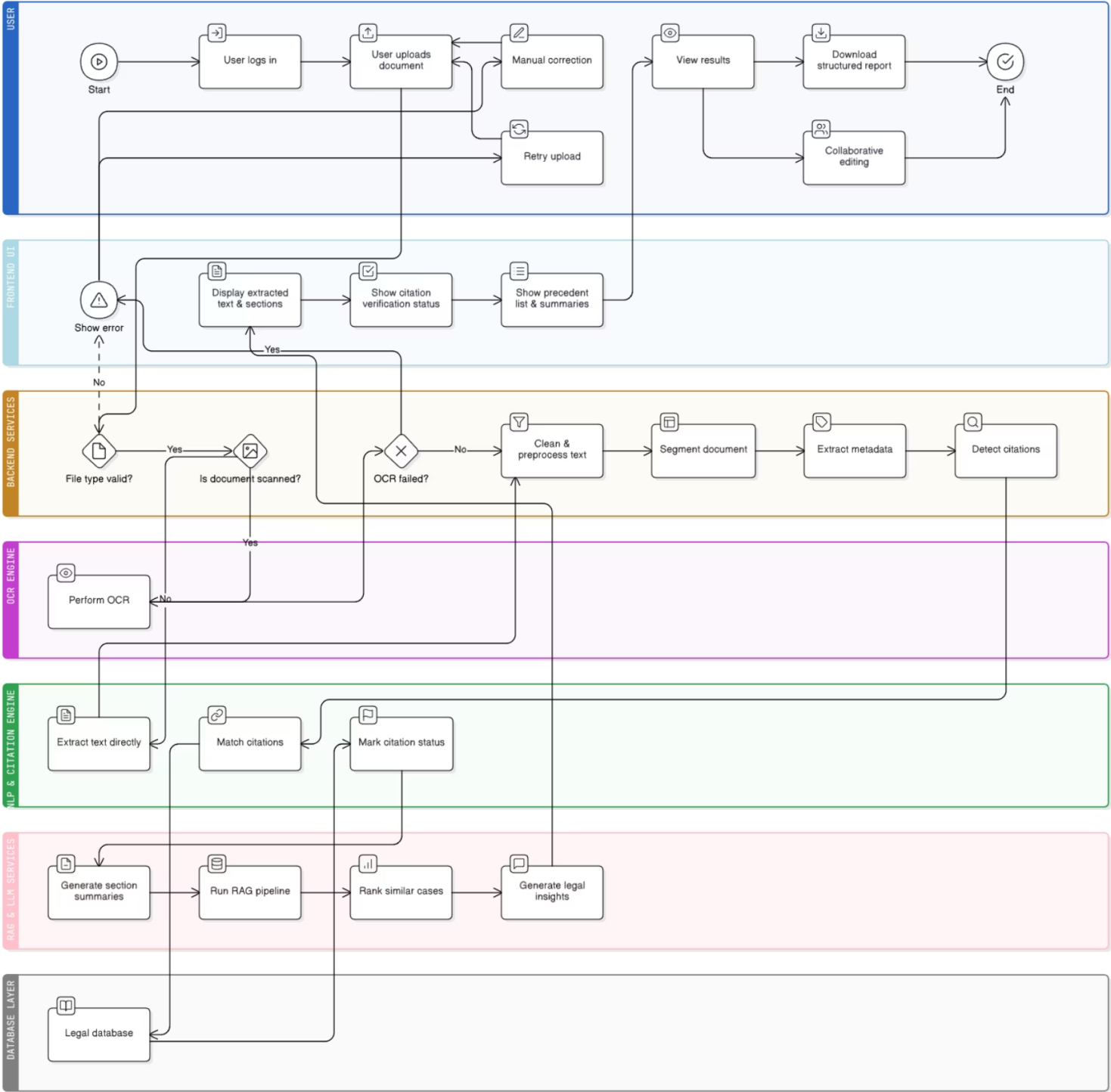
- **Frontend:** React + MUI + Tailwind for responsive web UI.
- **Backend:** FastAPI (Python) for APIs, NLP, OCR integration.
- **Database:** PostgreSQL for structured data; ElasticSearch for semantic search.
- **AI Layer:** Legal-BERT and Tesseract OCR integrated via API.
- **Deployment:** Dockerized, hosted on Heroku/Render (backend) & Vercel/Netlify (frontend).

Key Design: Modular, scalable, and maintainable system separating UI, business logic, and data layers.

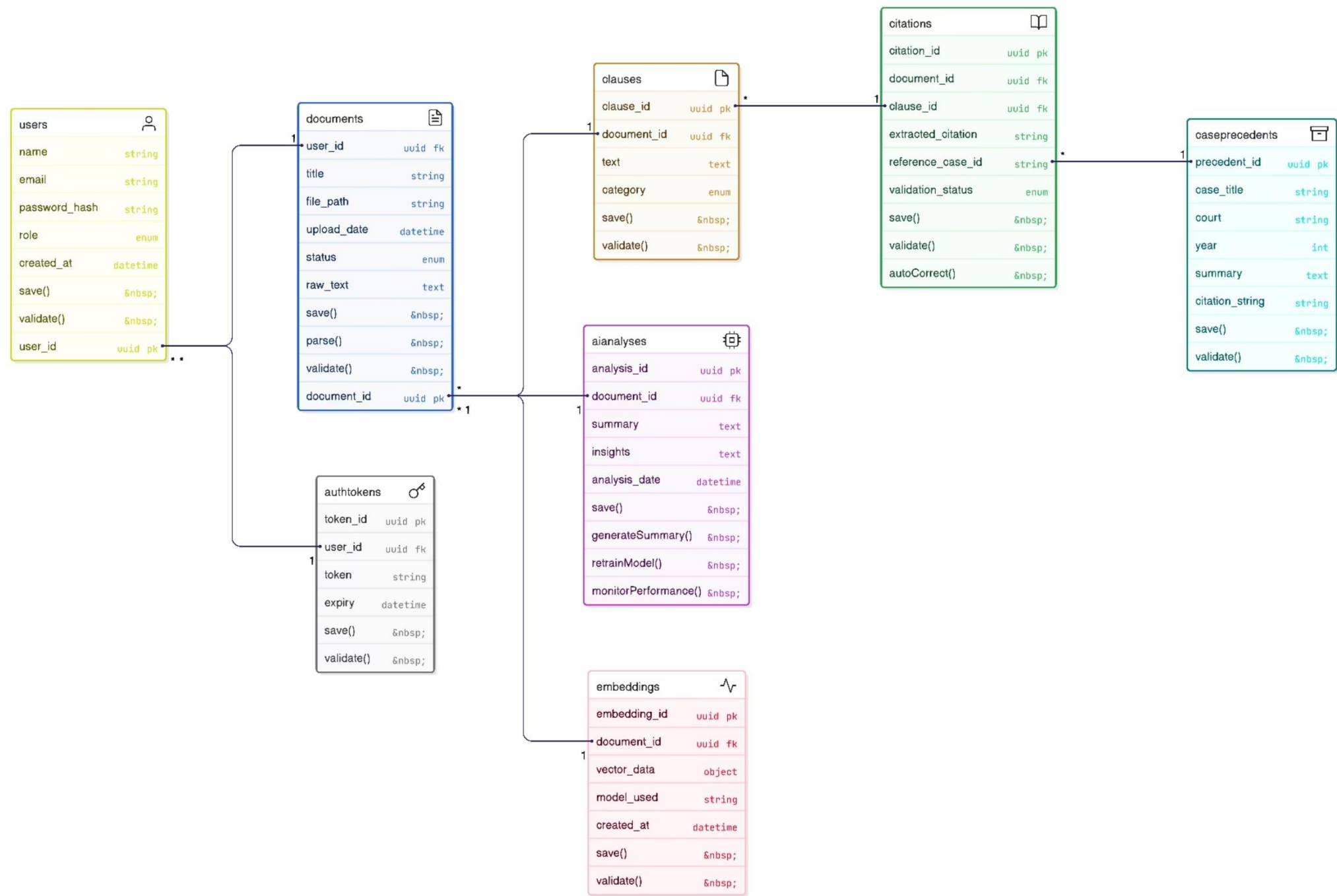
SYSTEM ARCHITECTURE DIAGRAM



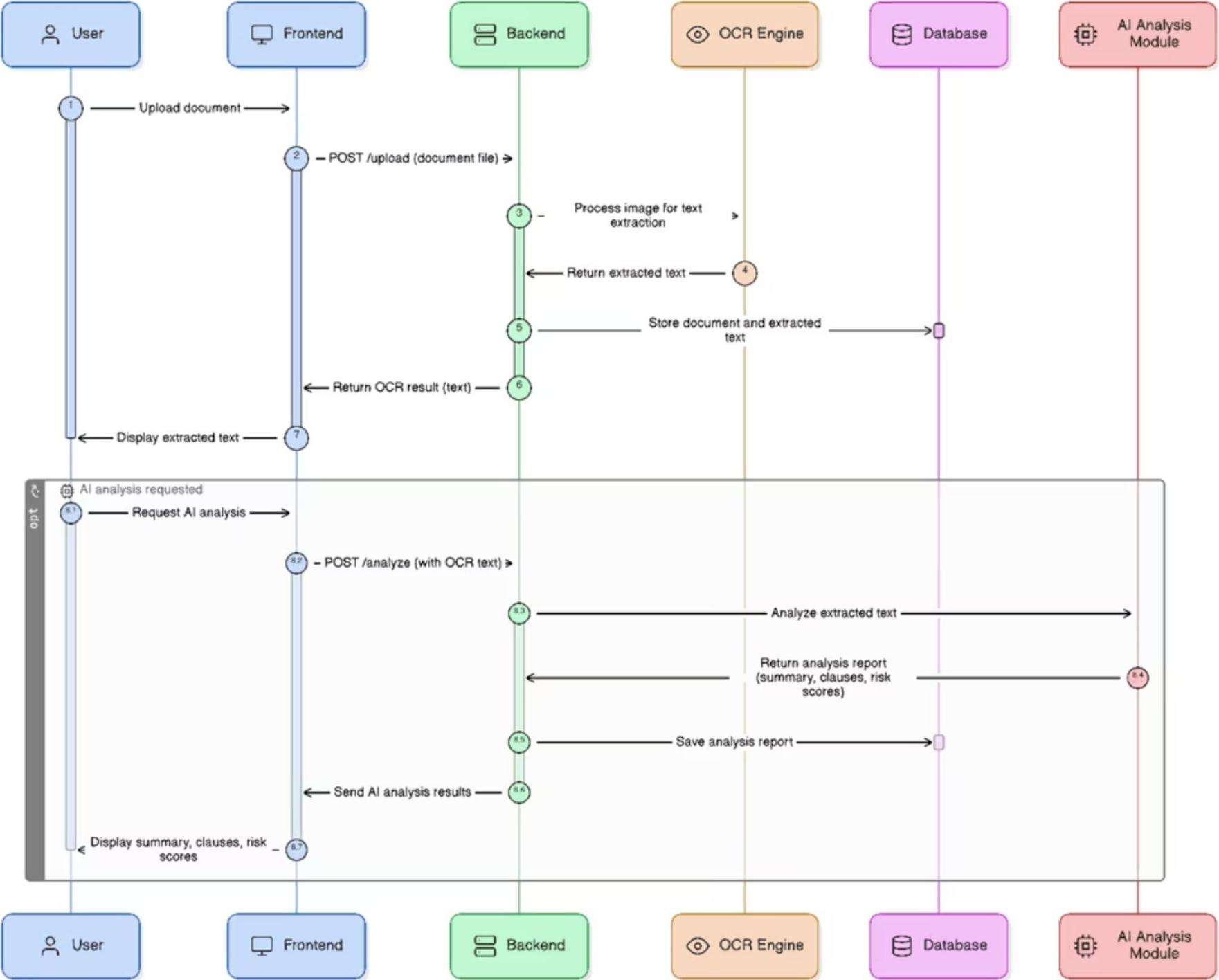
ACTIVITY DIAGRAM



CLASS DIAGRAM



SEQUENCE DIAGRAM



Data Design Overview

Users Table	Stores credentials, roles (Lawyer, Researcher, Student).
Documents Table	Stores metadata, OCR text, and extracted entities.
Citations Table	Holds citation references and validation results.
Search Index	ElasticSearch for semantic retrieval.
Embeddings	Vector representations stored in FAISS/Pinecone.

Storage:

- PostgreSQL → Structured data
- ElasticSearch for semantic search
- Redis → Cache for NLP and sessions

Iteration 1 Progress (Module 1)

Module: Document Ingestion and Preprocessing



Web Upload

Web-based document upload (React.js + Axios).



OCR Integration

Tesseract + OpenCV preprocessing.



Text Pipeline

Automatic text extraction and cleaning pipeline.



Backend API

API for OCR processing and response handling.



Testing & Accuracy

Unit and integration testing with 95%+ OCR accuracy on clean scans.

Next Steps:

- Integrate citation detection and storage in database.
- Extend preprocessing for legal documents.

Live Demo

Live Module: OCR-powered document ingestion with UI and backend connectivity.

Demonstration Includes:

1. User login and document upload.
2. OCR extraction and text display in frontend.
3. Process visualization (upload → preprocess → result).



Conclusion

Summary

- ADAL automates document analysis, enhancing accuracy and productivity in legal research.
- Core OCR and preprocessing module successfully implemented.
- Strong architectural base for next modules (citation verification, AI summarization).

Next Iteration Goals



NLP Integration

Integrate NLP-based citation validation and summarization.



Legal Assistant

Build the chatbot-based Legal Assistant module.



Testing

Prepare for full system integration and performance testing.